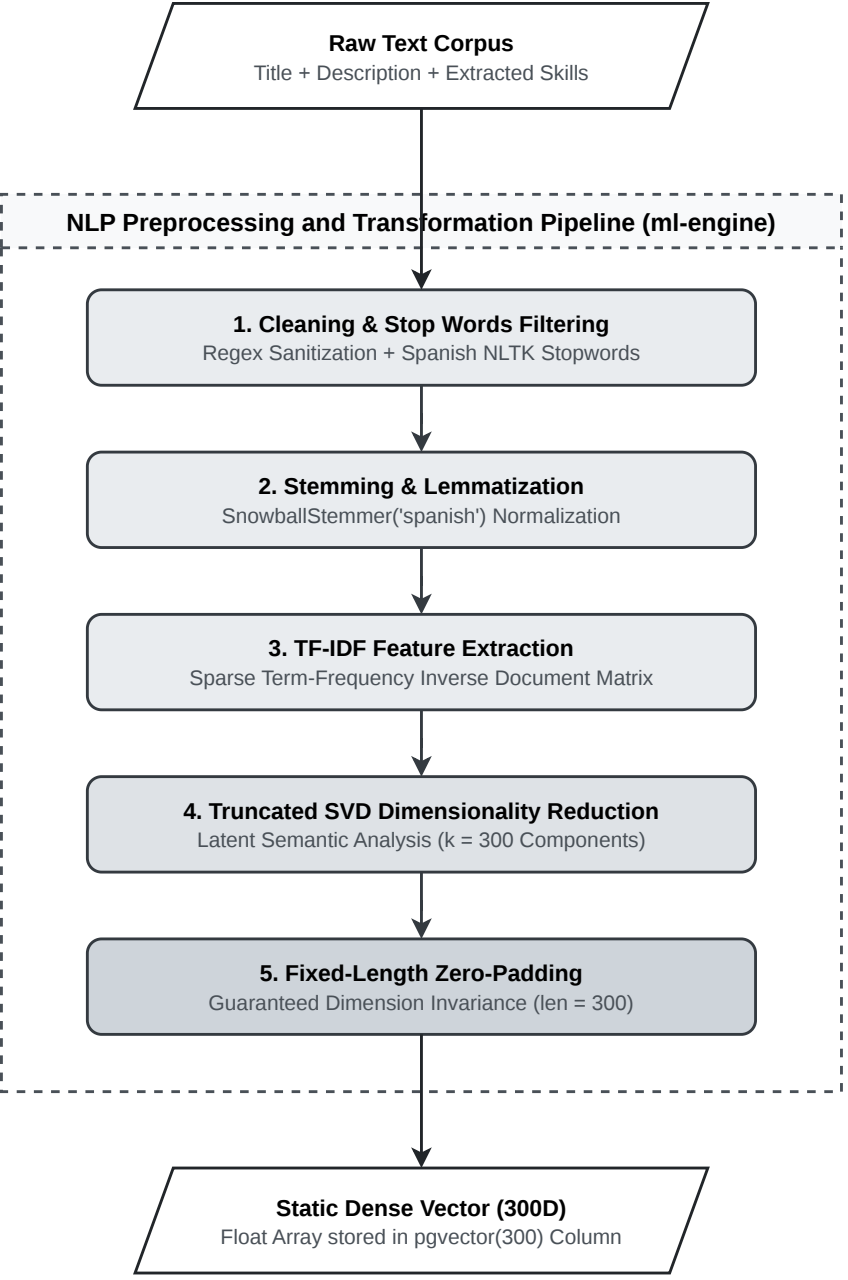
Raw Text Corpus
Title + Description + Extracted Skills
NLP Preprocessing and Transformation Pipeline (ml-engine)
1. Cleaning & Stop Words Filtering
Regex Sanitization + Spanish NLTK Stopwords
2. Stemming & Lemmatization
SnowballStemmer('spanish') Normalization
3. TF-IDF Feature Extraction
Sparse Term-Frequency Inverse Document Matrix
4. Truncated SVD Dimensionality Reduction
Latent Semantic Analysis (k = 300 Components)
5. Fixed-Length Zero-Padding
Guaranteed Dimension Invariance (len = 300)
Static Dense Vector (300D)
Float Array stored in pgvector(300) Column